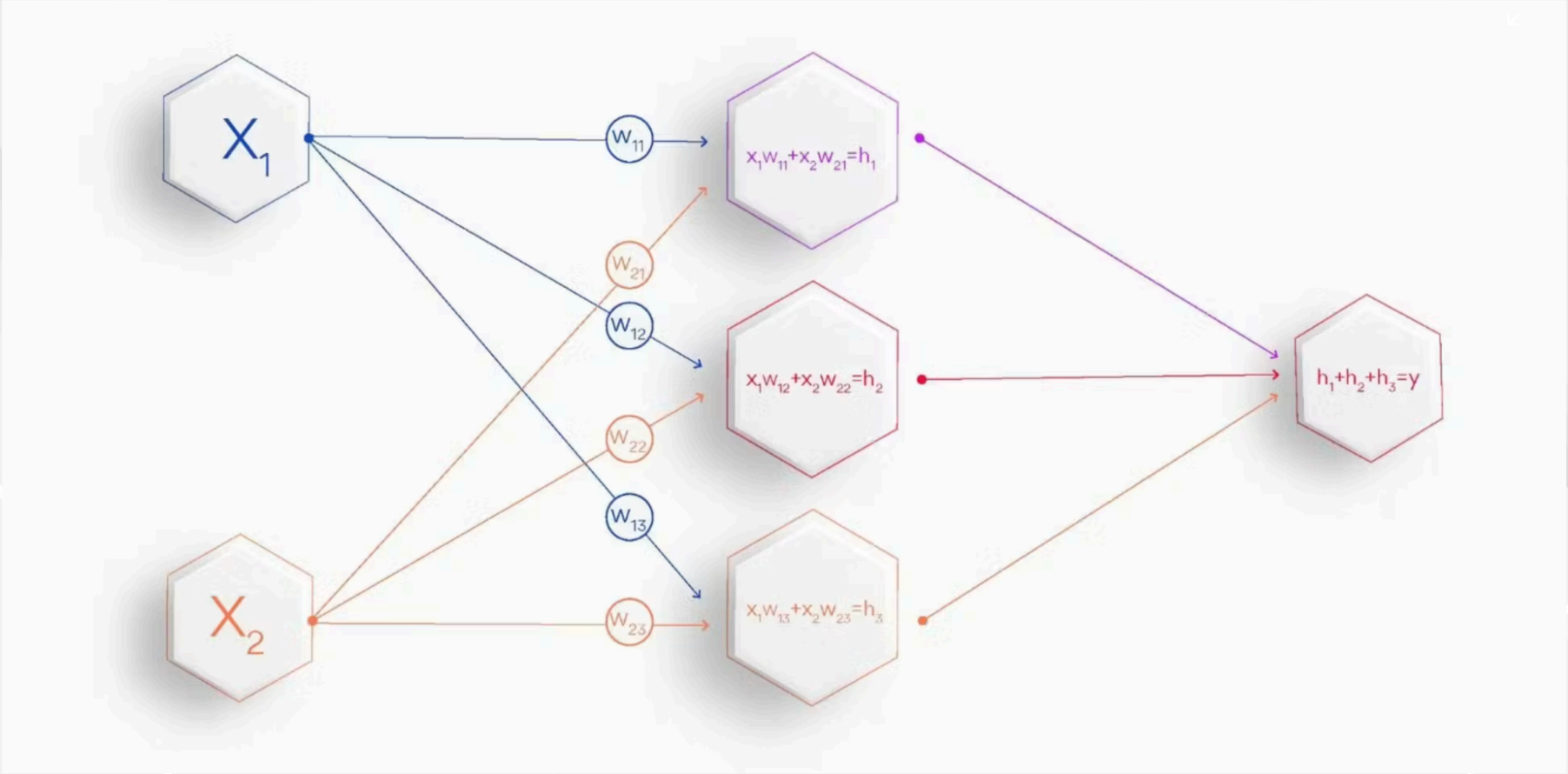
FFNN Architecture

Presented by: Brenden Park

OVERVIEW

- FFNN Architecture
- Different Types of Layers
- Pros
- Cons

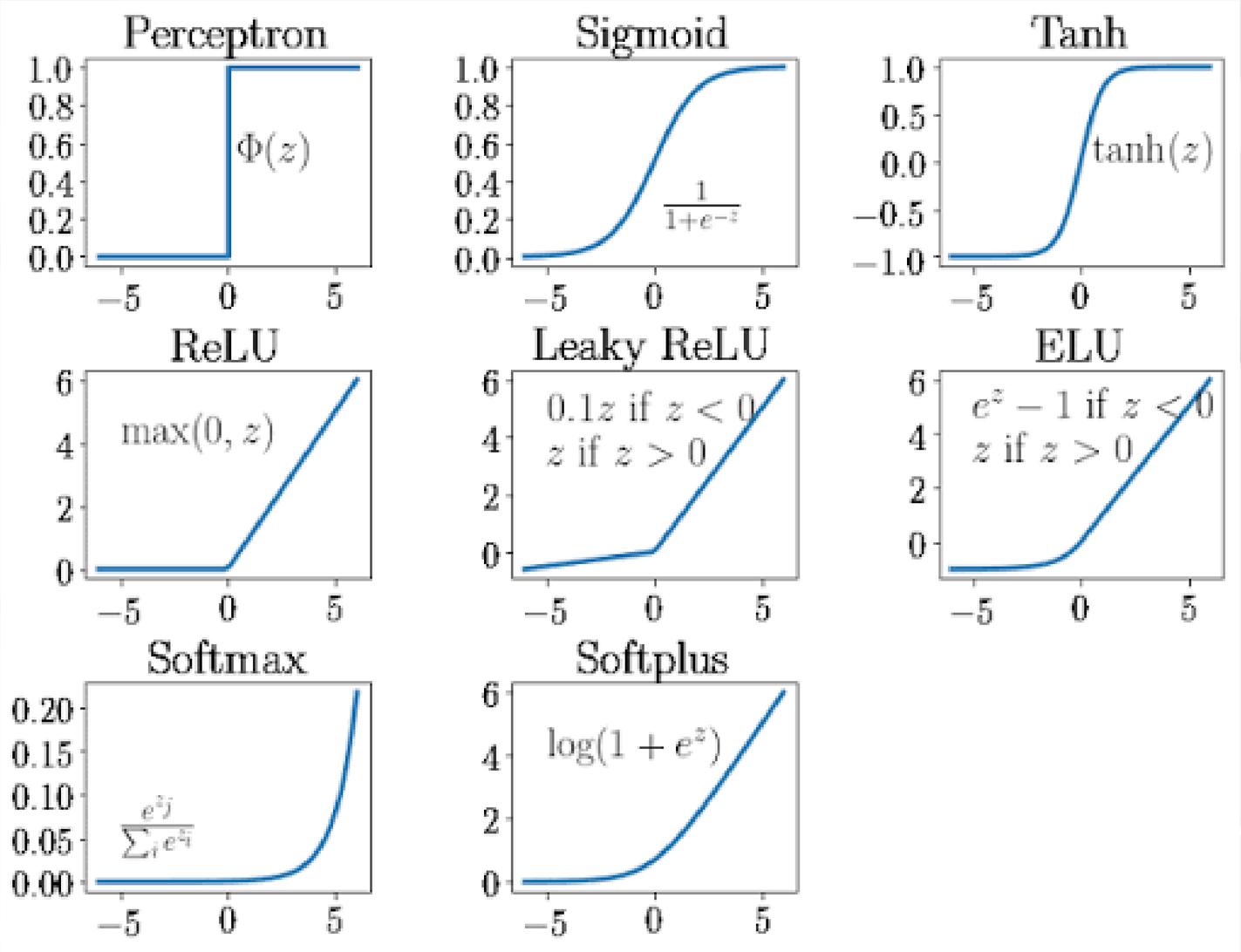
FFNN Architecture



Hidden Layer

$$[x_1 \quad x_2]\begin{bmatrix} w_{11} & w_{12} & w_{13} \\ w_{21} & w_{22} & w_{23} \end{bmatrix} = \begin{bmatrix} x_1 w_{11} + x_2 w_{21} \\ x_1 w_{12} + x_2 w_{22} \\ x_1 w_{13} + x_2 w_{23} \end{bmatrix}' = \begin{bmatrix} h_1 \\ h_2 \\ h_3 \end{bmatrix}'$$

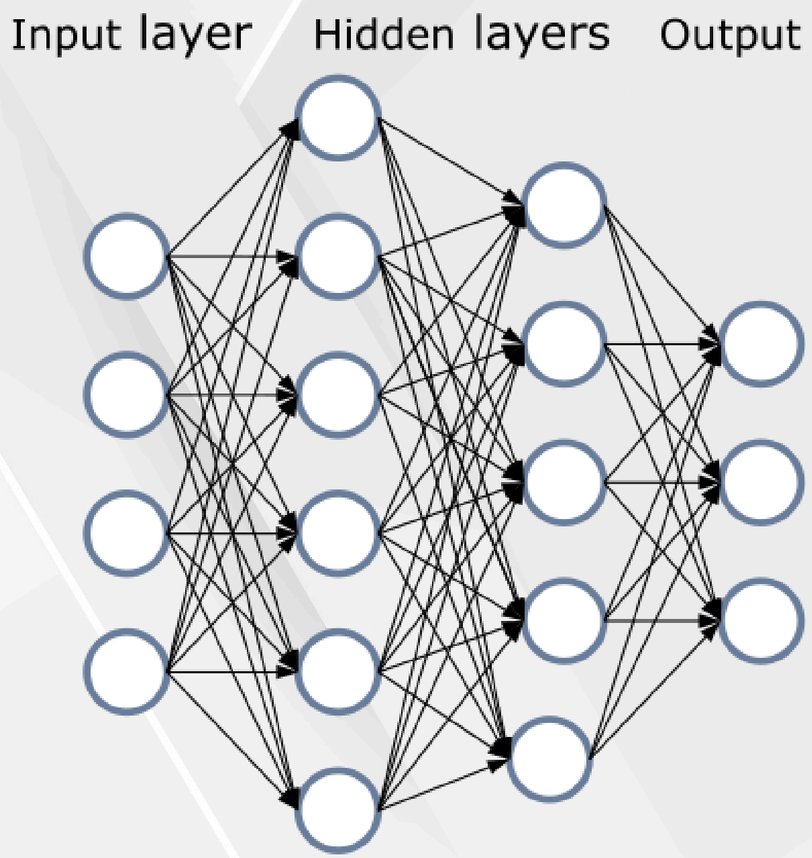
Activation Functions

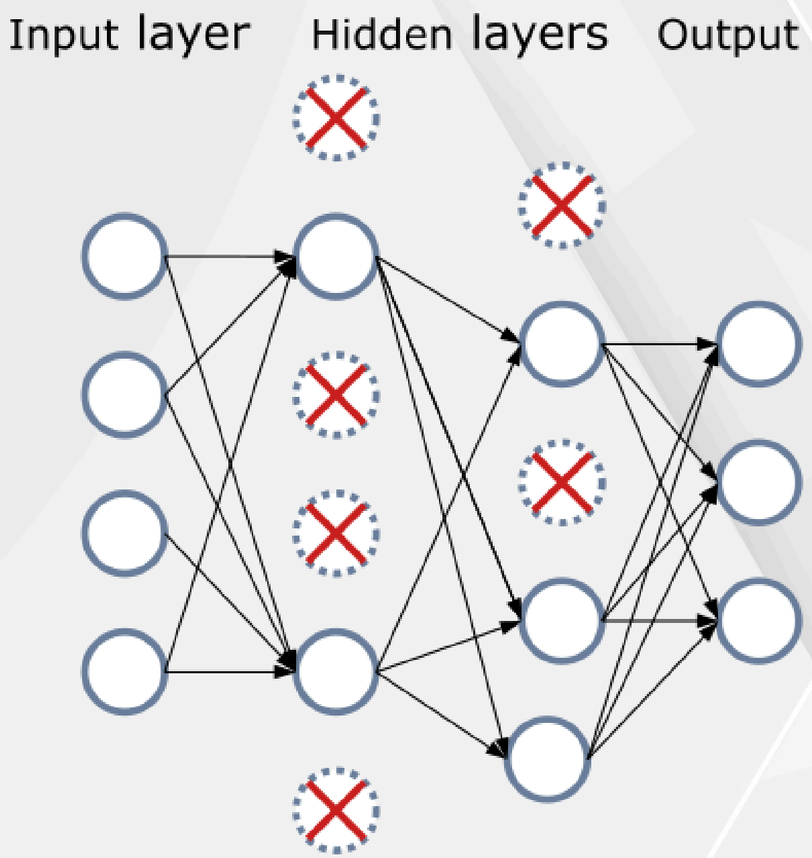


Output Layer

$$\begin{bmatrix} h_1 \\ h_2 \\ h_3 \end{bmatrix}' \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = h_1 + h_2 + h_3 = y$$

Dropout Layer





PROS

- Flexible Architecture
- Universal Theorem of Approximation
- Widely used for Classification Tasks

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SELFORMER: MOLECULAR REPRESENTATION LEARNING VIA SELFIES LANGUAGE MODELS

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ABSTRACT

Automated computational analysis of the vast chemical space is critical for numerous fields of research such as drug discovery and material science. Representation learning techniques have recently been employed with the primary objective of generating compact and informative numerical expressions of complex data, for efficient usage in subsequent prediction tasks. One approach to efficiently learn molecular representations is processing string-based notations of chemicals via natural language processing (NLP) algorithms. Majority of the methods proposed so far utilize SMILES notations for this purpose, which is the most extensively used string-based encoding for molecules. However, SMILES is associated with numerous problems related to validity and robustness, which may prevent the model from effectively uncovering the knowledge hidden in the data. In this study, we propose SELFormer, a transformer architecture-based chemical language model that utilizes a 100% valid, compact and expressive notation, SELFIES, as input, in order to learn flexible and high-quality molecular representations. SELFormer is pre-trained on two million drug-like compounds and fine-tuned for diverse molecular property prediction tasks. Our performance evaluation has revealed that, SELFormer outperforms all competing methods, including graph learning-based approaches and SMILES-based chemical language models, on predicting aqueous solubility of molecules and adverse drug reactions, while producing comparable results for the remaining tasks. We also visualized molecular representations learned by SELFormer via dimensionality reduction, which indicated that even the pre-trained model can discriminate molecules with differing structural properties. We shared SELFormer as a programmatic tool, together with its datasets and pre-trained models. Overall, our research demonstrates the benefit of using the SELFIES notations in the context of chemical language modeling and opens up new possibilities for the design and discovery of novel drug candidates with desired features.

Keywords Molecular representation learning · Drug discovery · Molecular property prediction · Natural language processing · Transformers

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Yuksel et. al, 2023
Two Linear Layers

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Molformer: Motif-Based Transformer on 3D Heterogeneous Molecular Graphs

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Abstract

Procuring expressive molecular representations underpins AI-driven molecule design and scientific discovery. The research mainly focuses on atom-level homogeneous molecular graphs, ignoring the rich information in subgraphs or motifs. However, it has been widely accepted that substructures play a dominant role in identifying and determining molecular properties. To address such issues, we formulate heterogeneous molecular graphs (HMGs), and introduce a novel architecture to exploit both molecular motifs and 3D geometry. Precisely, we extract functional groups as motifs for small molecules and employ reinforcement learning to adaptively select quaternary amino acids as motif candidates for proteins. Then HMGs are constructed with both atom-level and motif-level nodes. To better accommodate those HMGs, we introduce a variant of the Transformer named Molformer, which adopts a heterogeneous self-attention layer to distinguish the interactions between multi-level nodes. Besides, it is also coupled with a multi-scale mechanism to capture fine-grained local patterns with increasing contextual scales. An attentive farthest point sampling algorithm is also proposed to obtain the molecular representations. We validate Molformer across a broad range of domains, including quantum chemistry, physiology, and biophysics. Extensive experiments show that Molformer outperforms or achieves the comparable performance of several state-of-the-art baselines. Our work provides a promising way to utilize informative motifs from the perspective of multi-level graph construction. The code is available at https://github.com/smiles724/Molformer.

Introduction

The past decade has witnessed the extraordinary success of deep learning (DL) in many scientific areas. Inspired by these achievements, researchers have shown increasing interest in exploiting DL for drug discovery and material design with the hope of rapidly identifying desirable molecules. A key aspect of fast screening is how to represent molecules effectively, where graphs are a natural choice to preserve their internal structures. As a consequence, a number of *graph neural networks* (GNNs) (Gilmer et al. 2017; Ishida et al. 2021) have been invented and applied to molecular representation learning with noticeable performance.

However, most existing GNNs only take atom-level information in homogeneous molecular graphs as input, which fails to adequately exploit rich semantic information in motifs. Remarkably, motifs are significant subgraph patterns that frequently occur, and can be leveraged to uncover molecular properties. For instance, a carboxyl group (*COOH*) acts as a hydrogen-bond acceptor, which contributes to better stability and higher boiling points. Besides, similar to the role of N-gram in natural language, molecular motifs can promote the segmentation of atomic semantic meanings. While some regard motifs as additional features of atoms (Maziarka et al. 2020, 2021), these methods increase the difficulty to separate the semantic meanings of motifs from atoms explicitly and hinder models from viewing motifs from an integral perspective. Others (Huang et al. 2020) take motifs as the only input, but ignore the influence of single atoms and infrequent substructures.

To overcome these problems, we formulate a novel heterogeneous molecular graph (HMG) that is comprised of both atom-level and motif-level nodes as the model input. It provides a clean interface to incorporate nodes of different levels and prevents the error propagation caused by incorrect semantic segmentation of atoms. As for the determination of motifs, we adopt different strategies for different types of molecules. On the one hand, for small molecules, the motif lexicon is defined by functional groups based on chemical domain knowledge. On the other hand, for proteins that are constituted of sequential amino acids, a reinforcement learning (RL) motif mining technique is introduced to discover the most meaningful amino acid subsequences for downstream tasks.

In order to better align with HMGs, we present Molformer, an equivariant geometric model based on Transformer (Vaswani et al. 2017). Molformer differs from preceding Transformer-based models in two major aspects. First, it utilizes heterogeneous self-attention (HSA) to distinguish the interactions between nodes of different levels and incorporates them into the self-attention computation. Second, an Attentive Farthest Point Sampling (AFPS) algorithm is introduced to aggregate node features and obtain a comprehensive representation of the entire molecule.

To summarize, our contributions are as follows:

- To the best of our knowledge, we are the foremost to incorporate motifs and construct 3D heterogeneous molec-

[*]The corresponding author.
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Wu et. al, 2023
Two Dropout Layers w/ GELU

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BARTSMILES: GENERATIVE MASKED LANGUAGE MODELS FOR MOLECULAR REPRESENTATIONS

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ABSTRACT

We discover a robust self-supervised strategy tailored towards molecular representations for generative masked language models through a series of tailored, in-depth ablations. Using this pre-training strategy, we train `BARTSmiles`, a BART-like model with an order of magnitude more compute than previous self-supervised molecular representations. In-depth evaluations show that `BARTSmiles` consistently outperforms other self-supervised representations across classification, regression, and generation tasks setting a new state-of-the-art on 11 tasks. We then quantitatively show that when applied to the molecular domain, the BART objective learns representations that implicitly encode our downstream tasks of interest. For example, by selecting seven neurons from a frozen `BARTSmiles`, we can obtain a model having performance within two percentage points of the full fine-tuned model on task Clintox. Lastly, we show that standard attribution interpretability methods, when applied to `BARTSmiles`, highlight certain substructures that chemists use to explain specific properties of molecules. The code and the pretrained model are publicly available.

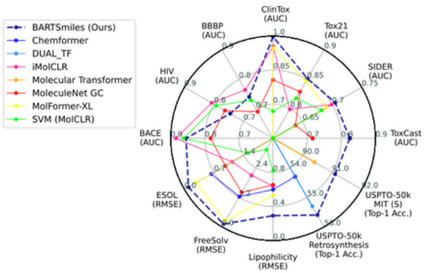


Figure 1: Performance of `BARTSmiles` on a variety of tasks compared to the current state-of-the-art.

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Chilingaryan et. al, 2022
One Linear Layer

CONS

- Highly Inefficient for Complex Tasks (Computationally Expensive)
- Overfitting

THANK YOU

QUESTIONS?